

Grade 6 Accelerated
Unit 13
Lesson 1 Homework

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1. What is the value of the expression $7 \cdot 4 - (x + 4)$ if $x = 2$?

$$\begin{aligned} &7 \cdot 4 - (2 + 4) \\ &7 \cdot 4 - 6 \\ &28 - 6 \\ &= 22 \end{aligned}$$

2. Find the value of $8 - 3(4 - y) + 2y \cdot 6$ when $y = 4$.

$$\begin{aligned} &8 - 3(4 - 4) + 2(4) \cdot 6 \\ &8 - 3(0) + 2(4) \cdot 6 \\ &8 - 0 + 8 \cdot 6 \\ &8 - 0 + 48 \\ &8 + 48 \\ &= 56 \end{aligned}$$

3. Evaluate $3p + \frac{6}{j} - 1$ for $p = 3$ and $j = 8$.

$$\begin{aligned} &3(3) + \frac{6}{8} - 1 \qquad \frac{78}{8} - \frac{8}{8} = \frac{70 \div 2}{8 \div 2} = \frac{35}{4} \text{ or } 8.75 \\ &9 + \frac{6}{8} - 1 \\ &\frac{72}{8} + \frac{6}{8} - 1 \\ &\frac{78}{8} - 1 \end{aligned}$$

4. Evaluate $10 - 20 \div 2f + 11$ if $f = 9$.

$$\begin{aligned} &10 - 20 \div 2(9) + 11 \\ &10 - 20 \div 2 \cdot 9 + 11 \\ &10 - 10 \cdot 9 + 11 \\ &10 - 90 + 11 \\ &-80 + 11 \\ &= -69 \end{aligned}$$

5. Find the value of $7 \cdot (m - 3m + 2)$ when $m = 1$.

$$\begin{aligned} &7 \cdot (1 - 3(1) + 2) \\ &7 \cdot (1 - 3 + 2) \\ &7 \cdot (-2 + 2) \\ &7 \cdot 0 \\ &= 0 \end{aligned}$$

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 4$ and $y = 3$.

$$\begin{aligned} 7 - (3 + 12)^2 + 6 \cdot 4 &= -218 + 24 \\ = 7 - 15^2 + 24 &= -194 \\ = 7 - 225 + 24 \end{aligned}$$


7. Find the value of $r^3 \div x \cdot c$ if $r = 4$, $x = 8$ and $c = 3$.

$$\begin{aligned} 4^3 \div 8 \cdot 3 \\ = 64 \div 8 \cdot 3 &= 24 \\ = 8 \cdot 3 \end{aligned}$$

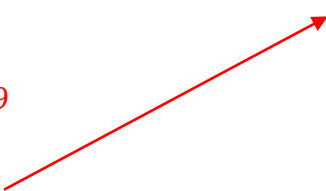
8. Evaluate the expression $3 \cdot (u^2 - 4) + \frac{k}{2}$ for $u = 2$ and $k = 6$.

$$\begin{aligned} 3 \cdot (2^2 - 4) + \frac{6}{2} &= 0 + 3 \\ = 3 \cdot (4 - 4) + 3 &= 3 \\ = 3 \cdot 0 + 3 \end{aligned}$$


9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 2$ and $h = 5$?

$$\begin{aligned} 5(2) \cdot 5^3 + (5 - 1)^2 &= 1250 + 16 \\ = 10 \cdot 125 + 4^2 &= 1,266 \\ = 10 \cdot 125 + 16 \end{aligned}$$


10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 4$ and $y = 7$?

$$\begin{aligned} \left(\frac{4^2-6}{4}\right) \cdot 7 + 7^2 &= 2.5 \cdot 7 + 49 \\ = \left(\frac{16-6}{4}\right) \cdot 7 + 49 &= 17.5 + 49 \\ = \frac{10}{4} \cdot 7 + 49 &= 66.5 \end{aligned}$$


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Lesson 2 Homework

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1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -3$.

$$\begin{aligned} & 8(-3) + 10 - (9 - 6)^2 \\ & -24 + 10 - 3^2 \\ & -24 + 10 - 9 \\ & -14 - 9 \\ & = -23 \end{aligned}$$

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = 8$, $q = -2$, and $s = -48$.

$$\begin{aligned} & -7(8) - 5(-2)^2 + \frac{-48}{2} \\ & -56 - 5(4) - 24 \\ & -56 - 20 - 24 \\ & -76 - 24 \\ & = -100 \end{aligned}$$

3. Evaluate $2s - 6q - f^2$ for $s = 4$, $q = -6$, and $f = -5$.

$$\begin{aligned} & 2(4) - 6(-6) - (-5)^2 \\ & 8 + 36 - 25 \\ & 44 - 25 \\ & = 19 \end{aligned}$$

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 3$, $d = 7$, $n = 9$, and $y = 1$.

$$\begin{aligned} & -3^3 - 3(7) + (9 - 1) \\ & -27 - 21 + 8 \\ & -48 + 8 \\ & = -40 \end{aligned}$$

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = 2$ and $z = 16$.

$$\begin{aligned} & 4(2)^4 - \frac{16}{4} \\ & 4(16) - 4 \\ & 64 - 4 \\ & = 60 \end{aligned}$$

6. What is the value of $5(8x + 4) - 3m^2$ if $x = -3$ and $m = 5$?

$$\begin{aligned} &5(8(-3) + 4) - 3(5)^2 \\ &5(-24 + 4) - 3(25) \\ &5(-20) - 3(25) \\ &-100 - 75 \\ &= -175 \end{aligned}$$

7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 8$, and $a = 7$.

$$\begin{aligned} &-9(8 - 6) + 2(7) - (9 - 7)^5 \\ &-9(2) + 2(7) - 2^5 \\ &-18 + 14 - 32 \\ &-4 - 32 \\ &= -36 \end{aligned}$$

8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = -4$, $k = 6$ and $y = 10$.

$$\begin{aligned} &(-4)^3 - 5 \cdot 6(10 - 6) \\ &-64 - 30(4) \\ &-64 - 120 \\ &= -184 \end{aligned}$$

9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = 2$.

$$\begin{aligned} &8 \div 2 - 12 \div (-10 + 4) \\ &8 \div 2 - 12 \div (-6) \\ &4 - (-2) \\ &= 6 \end{aligned}$$

10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 9$ and $a = 5$?

$$\begin{aligned} &4(9) - 10 + (5 - 11) \cdot 4^3 \\ &4(9) - 10 + (-6) \cdot 64 \\ &36 - 10 - 384 \\ &26 - 384 \\ &= -358 \end{aligned}$$

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Lesson 3 Homework

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Determine if each pair of expressions are equivalent.

1. $-9(20 + 7)$ and $-9(20) - 9(7)$
 $= 9(27)$ $- 180 - 63$
 $= -243$ $- 180 + -63$
 $= -243$
 $-243 = -243$
Equivalent

2. $4.5(10 + 7)$ and $4.5 \cdot 10 + 4.5 \cdot 7$
 $4.5(17)$ $45 + 31.5$
 $= 76.5$ $= 76.5$
 $76.5 = 76.5$
Equivalent

3. $\frac{3}{4} + \frac{9}{10}$ and $\frac{9}{10} + \frac{3}{4}$
 $\frac{30}{40} + \frac{36}{40}$ $\frac{36}{40} + \frac{30}{40}$
 $\frac{66}{40}$ $\frac{66}{40}$
 $= 1\frac{26}{40}$ $= 1\frac{26}{40}$
 $1\frac{13}{20} = 1\frac{13}{20}$
or
 $1.65 = 1.65$
Equivalent

4. $\frac{2}{3} - \frac{1}{2}$ and $\frac{1}{2} - \frac{2}{3}$
 $\frac{4}{6} - \frac{3}{6}$ $\frac{3}{6} - \frac{4}{6}$
 $\frac{1}{6} \neq -\frac{1}{6}$
They are not equivalent.

5. $\frac{1}{6} \times \frac{2}{9}$ and $\frac{2}{9} \times \frac{1}{6}$
 $\frac{1 \cdot 2}{6 \cdot 9}$ $\frac{2 \cdot 1}{9 \cdot 6}$
 $\frac{2}{54} = \frac{2}{54}$
Equivalent

6. $\frac{7}{11} \div \frac{1}{3}$ and $\frac{1}{3} \div \frac{7}{11}$
 $\frac{7}{11} \cdot \frac{3}{1}$ $\frac{1}{3} \cdot \frac{11}{7}$
 $\frac{21}{11}$ $= \frac{11}{21}$
 $= 1\frac{10}{11}$
 $1\frac{10}{11} \neq \frac{11}{21}$
They are not equivalent.

$$\begin{aligned}
 7. \quad & \frac{5}{7} - \left(\frac{1}{2} - \frac{3}{4} \right) \text{ and } \left(\frac{5}{7} - \frac{1}{2} \right) - \frac{3}{4} \\
 & \frac{5}{7} - \left(\frac{2}{4} - \frac{3}{4} \right) \quad \left(\frac{10}{14} - \frac{7}{14} \right) - \frac{3}{4} \\
 & \frac{5}{7} - \left(-\frac{1}{4} \right) \quad \frac{3}{14} - \frac{3}{4} \\
 & \frac{20}{28} + \frac{7}{28} \quad \frac{6}{28} - \frac{21}{28} \\
 & = \frac{27}{28} \quad = -\frac{15}{28}
 \end{aligned}$$

They are not equivalent.

$$\begin{aligned}
 8. \quad & \left(\frac{1}{5} \div \frac{2}{5} \right) \div \frac{8}{9} \text{ and } \frac{1}{5} \div \left(\frac{2}{5} \div \frac{8}{9} \right) \\
 & \frac{1}{5} \cdot \frac{5}{2} \cdot \frac{9}{8} \quad \frac{1}{5} \div \left(\frac{2}{5} \cdot \frac{9}{8} \right) \\
 & \frac{5}{10} \cdot \frac{9}{8} \quad \frac{1}{5} \div \left(\frac{18}{40} \right) \\
 & \frac{45}{80} \quad \frac{1}{5} \cdot \frac{40}{18} \\
 & \quad \quad \frac{40}{90} \\
 & = \frac{9}{16} \quad = \frac{4}{9}
 \end{aligned}$$

They are not equivalent.

9. Rewrite $(y - 11) + 26$ using the associative property.

$$y + (-11 + 26)$$

10. Rewrite $z + 14$ using the commutative property.

$$14 + z$$

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Lesson 4 Homework

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1. Rewrite the expression $(2z + 6) + 9z$ using the associative property.
 $2z + (6 + 9z)$
2. Rewrite the expression $(2z + 6) + 9z$ using the commutative property.
Answers can vary.
 $9z + (2z + 6)$
3. Rewrite the expression $5 \cdot (-3 \cdot 4g)$ using the associative property.
 $(5 \cdot -3) \cdot 4g$
4. Rewrite the expression $5 \cdot (-3 \cdot 4g)$ using the commutative property.
Answers can vary.
 $5 \cdot (4g \cdot -3)$
5. Rewrite the expression $-22r + 2(8k + 2)$ using the distributive property.
 $-22r + 16k + 4$
or
 $-22r + (2 \cdot 8k) + 2(2)$

6. Rewrite the expression $-22r + 2(8k + 2)$ using the commutative property of multiplication.

Answers may vary.

$$-r \cdot 22 + 2(8k + 2)$$

7. Rewrite the expression $-22r + 2(8k + 2)$ using the commutative property of addition.

Answers may vary.

$$-22r + 2(2 + 8k)$$

Write two equivalent algebraic expressions for each expression.

Answers can vary.

8. $(5 + 7) \cdot 3$
 $3(5) + 3(7)$
 $(7 + 5) \cdot 3$

9. $4t \cdot 3$
 $3 \cdot 4t$
 $4(t \cdot 3)$

10. $-5x + 2 - 10$
 $(-5x + 2) - 10$
 $-5x - 10 + 2$

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Lesson 5 Homework

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Write an algebraic expression for each written description.

1. The product of a number and seven.

$$7n$$

2. The quotient of a number and thirteen.

$$n \div 13 \text{ or } \frac{n}{13}$$

3. Two-thirds less than a number.

$$n - \frac{2}{3}$$

4. A number tripled.

$$3n$$

5. The difference of twenty-seven and a number.

$$27 - n$$

6. Gina will give each classmate, c , 5 free carnival ride tickets.

$$5c$$

7. There are 4 soda cases, each with s , number of cans in each case.

$$4s$$

8. A bag of candy has 250 calories per serving, s .

$$250s$$

9. A YouTube video pays \$0.01 per click, c , to the creator of the video.

$$0.01c$$

10. Frank scored 16 points less on the quiz than his classmate Jeannine, j .

$$j - 16$$

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Write an algebraic expression to represent each real-world situation.

1. Olivia has 15 videogames downloaded to her device. Her friend has x more videogames than her. How many songs does her friend have?

$$15 + x$$

2. Trevor has x tickets from a local arcade. His brother has 15 times more tickets than Trevor. How many tickets does his brother have?

$$15x$$

3. James earned 60 points while playing a card game. Arthur earned x fewer points than James. How many points did Arthur earn?

$$60 - x$$

4. Javier has 28 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?

$$28 \div x \text{ or } \frac{28}{x}$$

5. Francesca has a plant that measured h inches tall last week. This week, the plant is 12 inches tall.

$$12 - h$$

Answer may vary.

For each expression, create a real-world scenario that the algebraic expression could represent.

6. $x - 5$

Valerie had x dollars and spent \$5 on lunch.

7. $18x$

In a class of 18 students, each student earns x amount of prizes.

8. $\frac{150}{y}$

Jamie has 150 stickers that she wants to split evenly among y friends.

9. $19 + k$

Joshua had k dollars and earned \$19 More for mowing his neighbor's lawn.

10. $62 - h$

Tony has to complete 62 hours of community service and has completed h hours for helping at the food pantry.

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Lesson 7 Homework

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Write an equivalent expression by combining like terms.

1. $3x + 6x$

$$3x + 6x = 9x$$

2. $-4y + 8 + y - 5$

$$-4y + y + 8 - 5$$

$$-3y + 8 - 5$$

$$= -3y + 3$$

3. $-y + -y + y + y$

$$-1y + (-1y) + 1y + 1y$$

$$-2y + 2y$$

$$= 0$$

4. $7y - 2 + 6 - 2y$

$$7y - 2y + 6 - 2$$

$$= 5y + 4$$

5. $10 + y - 9 + y$

$$10 - 9 + y + y$$

$$= 1 + 2y$$

Find the sum of each expression.

$$\begin{aligned} 6. \quad & 2 + 9x - 1 + 5x = \\ & 9x + 5x + 2 - 1 \end{aligned}$$

$$= 14x + 1$$

$$\begin{aligned} 7. \quad & 4x + 1 + 10x - 8 = \\ & 4x + 10x + 1 - 8 \end{aligned}$$

$$= 14x - 7$$

$$\begin{aligned} 8. \quad & -3y + 1 + 2 + 6y = \\ & -3y + 6y + 1 + 2 \end{aligned}$$

$$= 3y + 3$$

$$\begin{aligned} 9. \quad & 10 + 2y - 8 + y = \\ & 10 - 8 + 2y + y \end{aligned}$$

$$= 2 + 3y$$

$$\begin{aligned} 10. \quad & 6y - 8 + y + 3 = \\ & 6y + y - 8 + 3 \end{aligned}$$

$$= 7y - 5$$

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Lesson 8 Homework

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Find the sum of each expression using the fewest terms possible.

1. $(0.3x + 0.6) + (0.7x - 0.8)$

$$0.3x + 0.7x + 0.6 - 0.8$$

$$= x - 0.2$$

2. $\left(-\frac{4}{5} + 8y\right) + \left(5y - \frac{1}{5}\right)$

$$8y + 5y - \frac{4}{5} - \frac{1}{5}$$

$$= 13y - 1$$

3. $(-0.5y - 2.4) + (y + 7.3)$

$$-0.5y + y - 2.4 + 7.3$$

$$= 0.5y + 4.9$$

4. $\left(7\frac{1}{2}y - \frac{2}{5}\right) + \left(6\frac{3}{5} - 2\frac{1}{4}y\right)$

$$7\frac{1}{2}y - 2\frac{1}{4}y - \frac{2}{5} + 6\frac{3}{5}$$

$$= 5\frac{1}{4}y + 6\frac{1}{5}$$

5. $\left(10 - \frac{1}{3}x\right) + \left(-9 + \frac{1}{3}x\right)$

$$-\frac{1}{3}x + \frac{1}{3}x + 10 - 9$$

$$= 1$$

Determine the value of each expression.

$$6. \quad (5.2 + 9.3x) + (-1.8x + 8.5) =$$

$$9.3x - 1.8x + 5.2 + 8.5$$

$$= 7.5x + 13.7$$

$$7. \quad \left(\frac{3}{8}x + \frac{1}{4}\right) + \left(\frac{3}{4}x - \frac{5}{16}\right) =$$

$$\frac{3}{8}x + \frac{3}{4}x + \frac{1}{4} - \frac{5}{16}$$

$$= 1\frac{1}{8}x - \frac{1}{16}$$

$$8. \quad (-3.9y + 5) + (3.9 + 6y) =$$

$$-3.9y + 6y + 5 + 3.9$$

$$= 2.1y + 8.9$$

$$9. \quad \left(1.0 + \frac{2}{9}y\right) + (-8.6 + y) =$$

$$\frac{2}{9}y + y + 1 - 8.6$$

$$= 1\frac{2}{9}y - 7.6$$

$$10. \quad \left(10.8 + \frac{2}{7}x\right) + \left(-\frac{9}{7} + x\right) =$$

$$\frac{2}{7}x + x + 10.8 - \frac{9}{7}$$

$$= 1\frac{2}{7}x + 9\frac{18}{35}$$

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Lesson 9 Homework

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Find the difference of each expression using the fewest terms possible.

1. $(0.4x + 0.9) - (0.5x - 0.7)$

$$0.4x - 0.5x + 0.9 + 0.7$$

$$\boxed{-0.1x + 1.6}$$

2. $\left(-\frac{3}{5} + 7y\right) - \left(4y - \frac{2}{5}\right)$

$$7y - 4y - \frac{3}{5} + \frac{2}{5}$$

$$\boxed{3y - \frac{1}{5}}$$

3. $(-0.4z + 1.9) - (z + 6.2)$

$$-0.4z - z + 1.9 - 6.2$$

$$\boxed{-1.4z - 4.3}$$

4. $\left(5\frac{1}{2}y - \frac{3}{5}\right) - \left(4\frac{1}{5} - 1\frac{3}{4}y\right)$

$$5\frac{1}{2}y + 1\frac{3}{4}y - \frac{3}{5} - 4\frac{1}{5}$$

$$\boxed{7\frac{1}{4}y - 4\frac{4}{5}}$$

5. $\left(10 - \frac{2}{9}q\right) - \left(-9 - \frac{2}{9}q\right)$

$$10 + 9 - \frac{2}{9}q + \frac{2}{9}q$$

$$\boxed{19}$$

Simplify each of the expressions.

6. $(8.2 - 5.6x) - (1.4x - 3.5) - (2.9x - 7.5)$

$$8.2 + 3.5 + 7.5 - 5.6x - 1.4x - 2.9x$$

$$19.2 - 9.9x$$

7. $\left(\frac{3}{8}x - \frac{1}{4}\right) + \left(\frac{1}{4}x + \frac{11}{16}\right) + \left(\frac{5}{8}x - \frac{3}{4}\right)$

$$\frac{3}{8}x + \frac{1}{4}x + \frac{5}{8}x - \frac{1}{4} + \frac{11}{16} - \frac{3}{4}$$

$$1\frac{1}{4}x - \frac{5}{16}$$

8. $(3.9y + 7) - (3.9 + 4y) + (-3.9 + 4y)$

$$3.9y - 4y + 4y + 7 - 3.9 - 3.9$$

$$3.9y - 0.8$$

9. $\left(1.8 + \frac{2}{9}r\right) + (-8.6 + r) - \left(-5.6 + \frac{1}{9}r\right)$

$$1.8 - 8.6 + 5.6 + \frac{2}{9}r + r - \frac{1}{9}r$$

$$-1.2 + 1\frac{1}{9}r$$

10. $\left(\frac{1}{6} + \frac{2}{7}n\right) - \left(-\frac{5}{6} + n\right) - \left(3\frac{1}{6} + 4\frac{4}{7}n\right)$

$$\frac{1}{6} + \frac{5}{6} - 3\frac{1}{6} + \frac{2}{7}n - n - 4\frac{4}{7}n$$

$$-2\frac{1}{6} - 5\frac{2}{7}n$$

Determine if the expressions are equivalent for the given values.

	Value	$7x + 3x - 6$	$2(5x - 8) + 10$	Equivalent?
1.	$x = -3$	$7(-3) + 3(-3) - 6$ $-21 + (-9) - 6$ $-30 - 6$ $= -36$	$2(5(-3) - 8) + 10$ $2(-15 - 8) + 10$ $2(-23) + 10$ $-46 + 10$ $= -36$	☒ yes ☐ no
2.	$x = 0$	$7(0) + 3(0) - 6$ $0 + 0 - 6$ $= -6$	$2(5(0) - 8) + 10$ $2(0 - 8) + 10$ $2(-8) + 10$ $-16 + 10$ $= -6$	☒ yes ☐ no
3.	$x = 0.3$	$7(0.3) + 3(0.3) - 6$ $2.1 + 0.9 - 6$ $3 - 6$ $= -3$	$2(5(0.3) - 8) + 10$ $2(1.5 - 8) + 10$ $2(-6.5) + 10$ $-13 + 10$ $= -3$	☒ yes ☐ no
4.	$x = \frac{1}{2}$	$7\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right) - 6$ $\frac{7}{2} + \frac{3}{2} - 6$ $\frac{10}{2} - 6$ $5 - 6$ $= -1$	$2\left(5\left(\frac{1}{2}\right) - 8\right) + 10$ $2(2.5 - 8) + 10$ $2(-5.5) + 10$ $-11 + 10$ $= -1$	☒ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

☒ always
☐ sometimes
☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{3}(9 + 6y) + 4x$	$2\left(2x + y + \frac{3}{2}\right)$	Equivalent?
6.	$x = -2$ $y = 6$	$\frac{1}{3}(9 + 6 \cdot 6) + 4(-2)$ $\frac{1}{3}(9 + 36) - 8$ $\frac{1}{3}(45) - 8 \rightarrow \frac{45}{3} - 8$ $15 - 8 = 7$	$2\left(2(-2) + 6 + \frac{3}{2}\right)$ $2\left(-4 + 6 + \frac{3}{2}\right)$ $2\left(2 + \frac{3}{2}\right)$ $2(3.5) = 7$	☐ yes ☐ no
7.	$x = 0$ $y = 1$	$\frac{1}{3}(9 + 6 \cdot 1) + 4(0)$ $\frac{1}{3}(9 + 6) + 0$ $\frac{1}{3}\left(\frac{15}{1}\right)$ $= 5$	$2\left(2 \cdot 0 + 1 + \frac{3}{2}\right)$ $2\left(1 + \frac{3}{2}\right)$ $2(2.5)$ $= 5$	☐ yes ☐ no
8.	Answers will vary $x = 2$ $y = 0$	$\frac{1}{3}(9 + 6 \cdot 0) + 4(2)$ $\frac{1}{3}\left(\frac{9}{1}\right) + 8$ $3 + 8 = 11$	$2\left(2 \cdot 2 + 0 + \frac{3}{2}\right)$ $2\left(4 + \frac{3}{2}\right)$ $2(5.5) = 11$	☐ yes ☐ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(9 - 11) - \frac{3}{5}(10 + 15x)$ $-2x \quad -6 - 9$ $-2x - 15$	$-11x - 15$	☐ yes ☐ no
10.	$\frac{5}{9}(6x + 12x) - (25 + 3x)$ $\frac{5}{9}(18x) - 25 - 3x$ $10x - 25 - 3x$ $7x - 25$	$-25 + 10x$	☐ yes ☐ no

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From the specified set of integers, determine which, if any, make the equation true.

$\{-4, -1, 1, 8\}$

- | | |
|---|--|
| <p>1. $12 + 5w = 6w + 4$
$12 + 5(-4) = 6(-4) + 4$
$12 - 20 = -24 + 4$
$-8 \neq -20$ NOT TRUE</p> <p>$12 + 5(-1) = 6(-1) + 4$
$12 - 5 = -6 + 4$
$7 \neq -2$ NOT TRUE</p> <p>$12 + 5(1) = 6(1) + 4$
$12 + 5 = 6 + 4$
$17 \neq 10$ NOT TRUE</p> <p>$12 + 5(8) = 6(8) + 4$
$12 + 40 = 48 + 4$
$52 = 52$ TRUE</p> | <p>2. $3 - x = 2x + 6$
$3 - (-4) = 2(-4) + 6$
$3 + 4 = -8 + 6$
$7 \neq -2$ NOT TRUE</p> <p>$3 - (-1) = 2(-1) + 6$
$3 + 1 = -2 + 6$
$4 = 4$ TRUE</p> <p>$3 - (1) = 2(1) + 6$
$3 - 1 = 2 + 6$
$2 \neq 8$ NOT TRUE</p> <p>$3 - (8) = 2(8) + 6$
$3 - 8 = 16 + 6$
$-5 \neq 22$ NOT TRUE</p> |
| <p>3. $3y - 5 = 8y + 15$
$3(-4) - 5 = 8(-4) + 15$
$-12 - 5 = -32 + 15$
$-17 = -17$ TRUE</p> <p>$3(-1) - 5 = 8(-1) + 15$
$-3 - 5 = -8 + 15$
$-29 \neq -49$ NOT TRUE</p> <p>$3(1) - 5 = 8(1) + 15$
$3 - 5 = 8 + 15$
$-2 \neq 23$ NOT TRUE</p> <p>$3(8) - 5 = 8(8) + 15$
$24 - 5 = 64 + 15$
$19 \neq 79$ NOT TRUE</p> | <p>4. $2u + 6 = 9u - 1$
$2(-4) + 6 = 9(-4) - 1$
$-8 + 6 = -36 - 1$
$-2 \neq -37$ NOT TRUE</p> <p>$2(-1) + 6 = 9(-1) - 1$
$-2 + 6 = -9 - 1$
$4 \neq -10$ NOT TRUE</p> <p>$2(1) + 6 = 9(1) - 1$
$2 + 6 = 9 - 1$
$8 = 8$ TRUE</p> <p>$2(8) + 6 = 9(8) - 1$
$16 + 6 = 72 - 1$
$22 \neq 71$ NOT TRUE</p> |

Determine if the inequality $18 \geq 2m - 6$ is true for each value.

5. 8

$$18 \geq 2(8) - 6$$

$$18 \geq 16 - 6$$

$$18 \geq 10$$

True

6. 12

$$18 \geq 2(12) - 6$$

$$18 \geq 24 - 6$$

$$18 \geq 18$$

True

7. 15

$$18 \geq 2(15) - 6$$

$$18 \geq 30 - 6$$

$$18 \geq 24$$

False

Determine if the inequality $-y + 3 + 8y \geq -3y - 7$ is true for each value.

8. -5

$$-(-5) + 3 + 8(-5) \geq -3(-5) - 7$$

$$5 + 3 - 40 \geq 15 - 7$$

$$-32 \geq 8$$

False

9. 0

$$3 + 7y \geq -3y - 7$$

$$3 + 7(0) \geq -3(0) - 7$$

$$3 \geq -7$$

True

10. 5

$$3 + 7y \geq -3y - 7$$

$$3 + 7(5) \geq -3(5) - 7$$

$$3 + 35 \geq -15 - 7$$

$$38 \geq -22$$

True

Grade 6 Accelerated
Unit 13
Lesson 12 Homework

Name Answer Key
Date _____
Period _____

Write an equivalent expression using the associative, commutative, or distributive property. State the property used. **Answers may vary**

1. $7x - \frac{3}{7}(14x - 21)$

$7x - 6x + 9$

distributive property

2. $(16x + 13) - 2x + 5$

$16x - 2x + 13 + 5$

commutative property

3. $\left(-\frac{2}{3}y + 8\right) \cdot 9$

$\left(8 - \frac{2}{3}y\right) \cdot 9$

commutative property

4. $-8\left(\frac{3}{4}x + 2\right) + (4 - 2x)$

$-6x - 16 + (4 - 2x)$

distributive property

5. If expressions are equivalent, then they will

- ☒ always
- ☐ sometimes
- ☐ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.7x - 0.1(4x - 2)$ and $7x - (4x - 2)$

$$0.7x - 0.4x + 0.2 \text{ and } 3x + 2$$

$$0.3x + 0.2$$

$$3(2) + 2$$

$$0.3(2) + 0.2$$

$$6 + 2$$

$$0.6 + 0.2$$

$$= 8$$

$$= 0.8$$

not equivalent

7. $(5x + 12) - 6$ and $7x - 2(x - 3)$

$$5x + 6$$

$$7x - 2x + 6$$

$$5x + 6$$

$$5(2) + 6$$

$$5(2) + 6$$

$$16$$

$$16$$

equivalent

8. $2\frac{1}{5}x - \frac{3}{10} - x$ and $\frac{6}{5}\left(x - \frac{2}{5}\right)$

$$\frac{11}{5}x - \frac{3}{10} - \frac{5}{5}x$$

$$\frac{6}{5}x - \frac{12}{25}$$

$$\frac{6}{5}x - \frac{3}{10}$$

$$\frac{6}{5} \cdot 5 - \frac{12}{25}$$

$$\frac{6}{5} \cdot 5 - \frac{3}{10}$$

$$6 - \frac{12}{25}$$

$$6 - \frac{3}{10} = 5\frac{7}{10}$$

$$5\frac{13}{25}$$

not equivalent

9. $0.2 + 0.1(14x - 2)$ and $2x - \left(\frac{3}{5}x - \frac{1}{5}\right)$

$$0.2 + 1.4x - 0.2$$

$$2x - \frac{3}{5}x + \frac{1}{5}$$

$$1.4x$$

$$\frac{10}{5}x - \frac{3}{5}x + \frac{1}{5}$$

$$\frac{7}{5}x + \frac{1}{5}$$

$$1.4(2) = 2.8$$

$$\frac{7}{5} \cdot 2 + \frac{1}{5} \rightarrow \frac{14}{5} + \frac{1}{5} = 3$$

not equivalent

10. Expressions that are equal are also equivalent.

☒ True

☐ False